Scratch: Animation, Costumes and Sound

Make characters walk, talk, change colour and play music.

Scratch | Beginner | Ages 6–12

The Big Idea

- Animation is flipping quickly through pictures. In Scratch each picture is a costume.
- The Costumes tab lets you draw, edit or upload costumes. Each sprite can have many.
- Looks blocks (purple) change how a sprite appears. Sound blocks (pink) play audio.
- Wait blocks control the speed. A short wait makes fast animation, a long wait makes slow animation.

Looks Blocks

- `next costume` switches to the next picture, `switch costume to [name]` picks one.
- `say [Hello!] for (2) seconds` and `think [Hmm] for (2) seconds` show speech bubbles.
- `show` and `hide` make a sprite appear or disappear.
- `set size to (100) %` and `change size by (10)` grow or shrink a sprite.
- `change [color] effect by (25)` shifts the colour. `clear graphic effects` resets it.
- `switch backdrop to [name]` changes the background.

Sound Blocks

- Open the Sounds tab to record, upload or choose a sound.
- `start sound [Meow]` plays a sound and moves on straight away.
- `play sound [Meow] until done` waits for the sound to finish.
- `set volume to (50) %` changes the loudness.
- Sounds in a `forever` loop with `play sound until done` make looping background music.

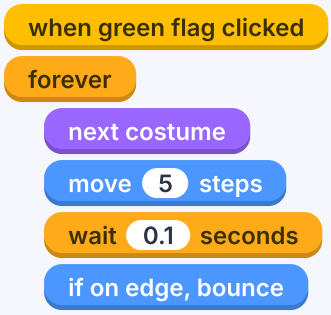
Animation Recipe

- Step 1: Add two or more costumes that look slightly different.
- Step 2: Put `next costume` inside a loop.
- Step 3: Add `wait (0.1) seconds` to set the speed.
- Step 4: Combine with a move block so the sprite also travels.

💻 Script Examples

Walking animation

Scratch blocks



▶ What happens:

The sprite flips between costumes while it walks, so the legs look like they are moving.

Colour cycle

Scratch blocks

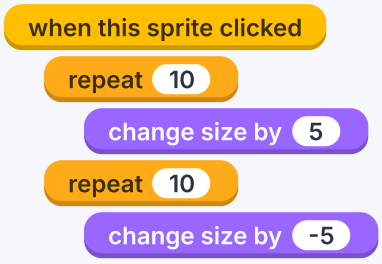


▶ What happens:

The sprite changes colour again and again like a rainbow.

Grow and shrink

Scratch blocks

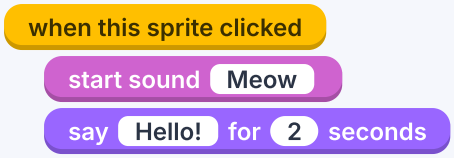


▶ What happens:

Clicking the sprite makes it swell up and shrink back to normal size.

Play a sound when clicked

Scratch blocks

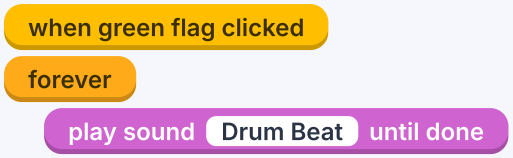


▶ What happens:

The sprite meows and says Hello! at the same time.

Background music

Scratch blocks



▶ What happens:

The sound plays again and again, so the music never stops.

Change the scene

Scratch blocks



▶ What happens:

Another sprite broadcasts night, and the backdrop changes to a starry sky.

📚 Key Terms

Costume
One picture of a sprite.

Volume
How loud a sound is.

Animation
Showing pictures one after another quickly.

Loop
Blocks that repeat.

Backdrop
The picture behind the sprites on the stage.

Wait
A pause that controls speed.

Effect
A visual change like colour or ghost.

Broadcast
A message that tells sprites to act.

Speech bubble
What say and think show.

Graphic effect
Colour, brightness, fisheye and other filters.

🧮 Quick Reference

```
next costume
switch costume to [costume2]
say [Hello!] for (2) seconds
change [color] effect by (25)
set size to (100) %
start sound [Meow]
play sound [Meow] until done
switch backdrop to [Stars]
```

✏️ Worked Examples

Q: Make a cat that looks like it is walking.

A: Use a sprite with two costumes. Put next costume, move (5) steps and wait (0.1) seconds inside a forever loop.

Q: Play a pop sound every time the sprite is clicked.

A: Add when this sprite clicked, then start sound [Pop]. Choose the sound from the Sounds tab first.

Q: Slow down an animation that is too fast.

A: Make the wait block longer. Change wait (0.1) seconds to wait (0.3) seconds.

🚫 Common Mistakes

- Forgetting `wait`, so the animation flickers too fast to see.
- Having only one costume, so `next costume` seems to do nothing.
- Using `start sound` in a forever loop, which starts many copies of the sound at once.
- Forgetting to reset size or colour effects when the project restarts.
- Making huge sounds files that slow the project down.

👨‍👩‍👧 Parent Corner

- Ask your child to draw a three picture flipbook on paper, then make it in Scratch.
- Ask which costume comes next, and what happens if you remove one.
- Encourage recording their own voice in the Sounds tab.
- Talk about frame rate: movies use many pictures per second too.
- Ask them to make a short animated story with two characters.

🎯 Quick Check

1. What is a costume?

✅ One picture of a sprite.

2. Which block makes animation?

✅ next costume inside a loop.

3. What does a wait block do in animation?

✅ It controls the speed.

4. Which block plays a sound and waits for it to finish?

✅ play sound until done.

5. How do you change the background?

✅ switch backdrop to [name].